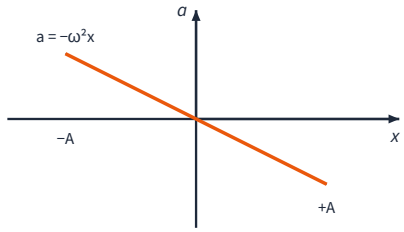
CHAPTER 20

# OSCILLATIONS

**Key idea:** in SHM the acceleration is proportional to the displacement and directed towards equilibrium.

## 1 SIMPLE HARMONIC MOTION




$$a = -\omega^2 x \qquad F = -kx$$

- $a \propto x$, always towards the equilibrium point
- $\omega = 2\pi f = \dfrac{2\pi}{T}$
- $a$–$x$ graph: straight line, gradient $-\omega^2$

## 2 KINEMATICS OF SHM

$$x = A\sin\omega t$$

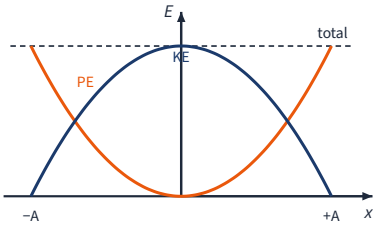
$$v = \omega A\cos\omega t$$

$$v = \pm\omega\sqrt{A^2 - x^2}$$

$$v_{\max} = \omega A \qquad a_{\max} = \omega^2 A$$

$x = A\cos\omega t$ if timing starts at an extreme.

## 3 ENERGY IN SHM




$$E = \tfrac{1}{2}m\omega^2 A^2 = \tfrac{1}{2}kA^2$$

$$E_k = \tfrac{1}{2}m\omega^2(A^2 - x^2)$$

$$E_p = \tfrac{1}{2}m\omega^2 x^2$$

Total energy $\propto$ (amplitude)$^2$; $E_k = E_p$ at $x = A/\sqrt{2}$.

## 4 SYSTEMS IN SHM

### MASS-SPRING

$$T = 2\pi\sqrt{\frac{m}{k}} \qquad \omega = \sqrt{\frac{k}{m}}$$

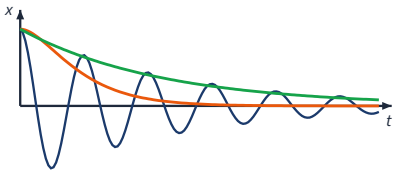
### SIMPLE PENDULUM (SMALL Θ)

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Independent of mass and amplitude; $\theta \lesssim 10^\circ$.

Springs: parallel $k = k_1 + k_2$; series $\dfrac{1}{k} = \dfrac{1}{k_1} + \dfrac{1}{k_2}$

## 5 DAMPED OSCILLATION

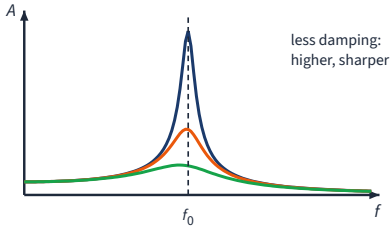



light (navy), critical (orange), heavy (green)

| | |
|---|---|
| **Light** | oscillates, amplitude decays exponentially |
| **Critical** | back to equilibrium fastest, no oscillation |
| **Heavy** | slow return, no oscillation |

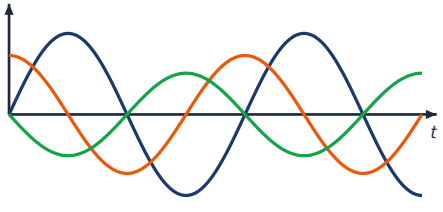
$$A = A_0 e^{-bt}$$

## 6 FORCED OSCILLATION & RESONANCE




- Forced: oscillates at the **driving** frequency
- Resonance: $f_{\text{driver}} = f_0$, maximum amplitude
- More damping: lower, broader peak, shifted slightly to lower $f$

## 7 PHASE RELATIONSHIPS




x (navy), v (orange), a (green)

$v$ leads $x$ by $\pi/2$; $a$ is in antiphase ($\pi$) with $x$.

## 8 COMMON SECTION A TRAPS

- Calculator in degrees for $\sin\omega t$: use **radians**.
- Saying the period depends on amplitude (it does not).
- Pendulum length: to the **centre** of the bob.
- Doubling $A$ doubles $v_{\max}$ but **quadruples** $E$.
- Speed is maximum at $x = 0$, acceleration at $x = \pm A$.

## EXAM TIPS

- ✓Prove SHM by showing $a = -(\text{constant})x$.
- ✓Identify $\omega$ as the coefficient of $t$.
- ✓Use energy for speed at a given $x$.
- ✓Sketch $x$, $v$, $a$ graphs one under another, $T/4$ apart.